Sepehr Khastoo

Junior Backend Developer, Python and Django

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Education

B.Sc. in Computer Engineering (Software) · Islamic Azad University, West Tehran Branch — Sep 2024 – present

Projects

Aegis Planner, Web · Creator, open source, v7.1.0 — 2026

`github.com/sepehrkhasto/aegis-zk-planner` · Django 4.2, Web Crypto API, Argon2, django-csp, GitHub Actions

- Built a zero-knowledge planner: the browser encrypts all data with AES-256-GCM, and the server sees only ciphertext.
- Implemented PBKDF2-SHA256 key wrapping (600,000 iterations): a password change re-wraps one key, not the vault.
- Added revision-checked saves that return HTTP 409 on stale writes, preventing silent overwrites between devices.
- Hardened the server with a strict CSP, Argon2 hashing, per-IP rate limiting and account lockout; 146 Django tests and 556 JavaScript checks run in CI.

Aegis Planner, Desktop · Creator, open source, v2.11.10 — 2026

`github.com/sepehrkhasto/aegis-planner` · Python, PyQt6, cryptography, pytest, PyInstaller

- Shipped an offline, encrypted planner for Windows with a Persian interface and a Jalali calendar.
- Made storage crash-safe with atomic writes (temporary file, fsync, rename) and encrypted backups with verify and restore.
- Set up release builds from tagged source in GitHub Actions, published with SHA-256 checksums; 971 pytest tests.

Refinery Risk Monitor · Project Lead, team of four — 2026

`github.com/sepehrkhasto/refinery-risk-monitor` · FastAPI, SQLAlchemy 2.0, Alembic, JWT, WebSocket, scikit-learn

- Led a team of four through scope, architecture and delivery of a FastAPI platform for live risk scoring and forecasting; placed in the top three of the course.
- Directed the backend, built with AI coding tools: JWT role-based access (operator, engineer, admin) on every endpoint; production refuses to start with a default secret key or wildcard CORS.
- The platform serves P10/P50/P90 forecasts and SHAP explanations over REST and streams live risk scores over WebSocket.

Andarun · Creator, in development — 2026 – present

`github.com/sepehrkhasto/andarun` · Python 3.12, Django 6.1

- Building an invite-only, text-only social network designed to run on a domestic network when international access is cut.
- Implemented a custom user model with username login and strict usernames (`[a-z0-9_]{3,20}`, stored in lowercase), with fail-closed settings: no default secret key, DEBUG off by default.

Skills

Languages	Python, SQL; C and C++ (basic)
Backend	Django (ORM, migrations, custom user model, admin, forms and validation); automated testing with Django TestCase; also used FastAPI, SQLAlchemy, Alembic and WebSocket
Databases	PostgreSQL, MySQL, SQLite; relational schema design
Security	Web Crypto API (AES-256-GCM, PBKDF2 key wrapping), Argon2, Content Security Policy, JWT role-based access, environment-based configuration with safe defaults
Tools	Git and GitHub (feature branches, conventional commits, fast-forward merges), GitHub Actions, PyQt6
Familiar with	Class-based views and mixins; authentication, sessions and permissions; ORM query optimization (N+1, select_related, prefetch_related); transactions and row-level locking; Docker; Linux command line; networking fundamentals (Network+)
Learning	Django REST Framework
Soft skills	Team leadership, technical writing, self-directed learning

Other experience

Logistics Operations Specialist, Baroline, Tehran · Jun – Aug 2026, full-time

Languages

Persian (native), English (B1)